

ADSC3910
Integrated Practice - 2

Thompson Rivers University
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Early Detection of Lung Cancer Risk Using Prediction Models

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1.Introduction

Lung cancer remains one of the most serious health challenges worldwide, responsible for more cancer-related deaths than any other type of cancer. Despite efforts to raise awareness about its risk factors, such as smoking, environmental pollutants, and genetic predisposition, lung cancer continues to be diagnosed at later stages when treatment options are limited and less effective. Early detection is critical in improving survival rates and patient outcomes, as patients diagnosed in the early stages of lung cancer have significantly higher chances of successful treatment. However, current diagnostic methods, including imaging and biopsy, are often expensive, invasive, and inaccessible.

To address this problem, the proposal seeks to leverage modern data science techniques to build a predictive model to identify individuals at high risk of developing lung cancer. By analyzing patient data such as demographics, medical history, lifestyle factors, and genetic information, the model will predict lung cancer's likelihood before symptoms become apparent. This data-driven approach offers a non-invasive, cost-effective way to enhance early detection efforts, potentially reducing the burden of late-stage diagnoses and improving patient outcomes.

The final data product for this project will deliver a comprehensive solution to support healthcare professionals and researchers in early lung cancer detection. The key components include:

1. **Predictive Model:** A machine learning model trained on patient data to predict lung cancer risk, providing probabilistic predictions to assist in identifying high-risk individuals for further screening. The model will be optimized for accuracy and interpretability.
2. **Data Pipeline:** A scalable pipeline for data cleaning, feature engineering, and transformation, ensuring seamless flow of patient data to the predictive model. It will be easily integrated into healthcare systems for real-time or batch processing.
3. **Technical Report:** A formal report summarizing the methodology, findings, and recommendations, including model performance metrics such as accuracy, sensitivity, and specificity. This will help guide future improvements and implementations.

2. Data Science Techniques

In this project, we will apply data science techniques to develop a predictive model that assesses a patient's likelihood of having lung cancer based on clinical and lifestyle variables. Using demographic and medical data, we will train and test machine learning models to identify the most

important factors contributing to lung cancer risk, ultimately improving early detection and intervention.

2.1 Appropriateness of the data

The dataset is well-suited for classification techniques, given the binary target variable (LUNG_CANCER: YES/NO) and the presence of categorical and binary features. Models like Logistic regression, Random Forest and XGBoost are ideal due to their ability to handle such data effectively. These features represent meaningful clinical and lifestyle factors, making it possible to uncover patterns relevant to lung cancer risk and develop accurate predictive models.

2.2 Difficulties due to data

The dataset may present challenges like imbalanced classes, where fewer lung cancer cases could skew predictions. We'll address this using techniques like oversampling or under sampling. For missing data, we'll apply mean/mode imputation or remove incomplete records based on the severity. Additionally, multicollinearity between features (e.g., coughing and shortness of breath) could lead to overfitting, so we'll monitor correlations and remove redundant features when necessary.

2.3 Data Fields

Data Field	Description
GENDER	Gender of the patient (M for Male, F for Female)
AGE	Age of the patient in years
SMOKING	Indicates smoking status (1 for Yes, 0 for No)
YELLOW_FINGERS	Indicates presence of yellow fingers (1 for Yes, 0 for No)
ANXIETY	Indicates presence of anxiety (1 for Yes, 0 for No)
PEER_PRESSURE	Indicates level of peer pressure (0 for None, 1 for Low, 2 for High)
CHRONIC_DISEASE	Indicates presence of chronic diseases (0 for None, 1 for Mild, 2 for Severe)
FATIGUE	Indicates presence of fatigue (1 for Yes, 0 for No)
ALLERGY	Indicates presence of allergies (0 for None, 1 for Mild, 2 for Severe)
WHEEZING	Indicates presence of wheezing (1 for Yes, 0 for No)
ALCOHOL_CONSUMING	Indicates alcohol consumption status (1 for Yes, 0 for No)
COUGHING	Indicates presence of coughing (1 for Yes, 0 for No)
SHORTNESS_OF_BREATH	Indicates presence of shortness of breath (1 for Yes, 0 for No)

Data Field	Description
SWALLOWING_DIFFICULTY	Indicates presence of swallowing difficulty (1 for Yes, 0 for No)
CHEST_PAIN	Indicates presence of chest pain (1 for Yes, 0 for No)
LUNG_CANCER	Indicates the diagnosis of lung cancer (YES for diagnosed, NO for not diagnosed)

2.4 Data Overview

GENDER	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	CHRONIC_DISEASE	FATIGUE	COUGHING	SHORTNESS_OF_BREATH	LUNG_CANCER
M	65	1	1	0	1	1	1	1	YES
F	58	0	0	1	2	1	1	0	NO
M	70	1	0	1	0	1	0	1	YES

The lung cancer JSON dataset contains a total of 3,000 documents.

2.5 Data pipeline and ML models

In this project, we will start by loading the JSON dataset into a structured format, like a Pandas Data Frame, to facilitate exploratory data analysis (EDA). We will summarize each feature using descriptive statistics and create visualizations, such as heat maps, to identify patterns and correlations, particularly focusing on missing or inconsistent data. Next, we will perform feature engineering by converting categorical variables into numerical formats through techniques like one-hot encoding, scaling numeric features, and handling any missing data through imputation or removal. Finally, we will build and evaluate several supervised learning models, including Logistic Regression, Random Forest, and XGBoost, to predict lung cancer diagnoses.

2.6 Evaluation metrics

Model performance will be assessed using key evaluation metrics:

1. **Accuracy:** Measures overall correctness of predictions, indicating how well the model identifies cases.
2. **Precision:** Evaluates the proportion of true positives among all positive predictions, minimizing false positives and ensuring only likely cancer patients are flagged.
3. **Recall:** Also known as sensitivity, it measures the proportion of true positives among actual cases, ensuring as many lung cancer cases as possible are detected.
4. **F1-Score:** The harmonic mean of precision and recall, providing a balanced metric that is crucial in healthcare to minimize both false positives and false negatives.

2.7 Success criteria

The success criteria include achieving at least 90% accuracy, 85% precision, 80% recall, and an F1-score of 0.75 or higher to ensure balanced model performance.

2.8 Other ADSC courses

Our project integrates concepts from ADSC 3610 - Database Systems in Applied Data Science II by utilizing MongoDB as our database management system:

The project will be underpinned by the use of various sophisticated tools such as MongoDB (for storage, retrieval and manipulation of data), Visual Studio Code (IDE), PyMongo distribution (for interaction between the MongoDB databases and IDE) and various python libraries.

3. Timeline

MILESTONE	PLANNED START DATE	PLANNED END DATE
1. Data Preprocessing (Querying Data, Handling Missing Data, Normalization/Scaling, Handling Outliers, Selecting Features, Removing Unnecessary Columns)	10/07/2024	10/14/2024
2. Data Pipeline Integration (Develop and Integrate Data Pipeline with MongoDB)	10/14/2024	10/21/2024
3. Model Development (Selecting Relevant Models, Data Train/Test Split, Comparing Accuracy, Selecting Best Model)	10/21/2024	10/28/2024
4. Documentation and Presentation (Presentation Material, Final Report Preparation)	10/28/2024	11/04/2024
5. Final Project Deliverables (Project Review, Final Changes, Submission)	11/04/2024	11/27/2024

4. References

<https://www.kaggle.com/datasets/akashnath29/lung-cancer-dataset?select=dataset.json>